

Marathon Training (run)

Luca has recently stepped up his training with the ultimate goal of being able to run a half marathon: as soon as he can, he goes out for a run. Since he always wears a fitness tracker, every time he trains the covered distance is automatically recorded in his favourite fitness application. So far, the details of N training sessions have been captured. Among other information, one can see that in the i -th session he ran for D_i kilometres.



Figure 1: Luca during one of his training sessions.

Luca wants to analyse the progress to check whether he is getting better. Specifically, he would like to know how big it is his longest *improving streak*. A sequence of consecutive training sessions, starting from the x -th one and ending at the y -th one, is an *improving streak* if Luca has run at least as many kilometres in the last session as in the first one and if he has covered at least D_x and at most D_y kilometres in every one of them (formally, $D_x \leq D_k \leq D_y$ for each $k = x, \dots, y$).

How long is the longest *improving streak* Luca has recorded?

 Among the attachments of this task you may find a template file `run.*` with a sample incomplete implementation.

Input

The first line contains the only integer N , the number of training sessions recorded by Luca. The second line contains N integers D_i , the distance covered in the i -th training session.

Output

You need to write a single line with an integer: the length of the longest *improving streak* of training sessions.

Constraints

- $1 \leq N \leq 100\,000$.
- $0 \leq D_i \leq 10^9$ for each $i = 0 \dots N - 1$.

Scoring

Your program will be tested against several test cases grouped in subtasks. In order to obtain the score of a subtask, your program needs to correctly solve all of its test cases.

- **Subtask 1** (0 points) Examples.

- **Subtask 2** (17 points) $D_i \leq 1$ for each $i = 0 \dots N - 1$.

- **Subtask 3** (20 points) $N \leq 500$.

- **Subtask 4** (26 points) $N \leq 10\,000$.

- **Subtask 5** (37 points) No additional limitations.


Examples

input	output
5 2 1 3 5 10	4
6 2 7 3 8 4 7	4

Explanation

In the **first sample case**, there is one *improving streak* of length four, consisting of the last four training sessions. In each of those four sessions Luca has run at least 1 and at most 10 kilometres. No longer improving streaks exist.

In the **second sample case**, there is one *improving streak* of length four, composed of the first four training sessions. In the first four sessions, Luca has always run at least 2 and at most 8 kilometres. No longer improving streaks exist.